MAGIVOX AI

JavaScript - A to Z

Core JS | DOM | Async | APIs | Modules | Debugging | Interview

How to use this guide
Study one topic, type the example yourself, practise without copying, then take the MAGIVOX topic test. After all modules, take the final course exam.

Learning outcomes

- JavaScript Foundation
- Variables and Types
- Operators and Conditions
- Loops
- Functions
- Arrays

Table of Contents

1. JavaScript Foundation
2. Variables and Types
3. Operators and Conditions
4. Loops
5. Functions
6. Arrays
7. Objects
8. DOM
9. Forms and Validation
10. Async JavaScript
11. Modules
12. Storage
13. Debugging
14. Project
15. Interview Preparation

1. JavaScript Foundation

What to learn

- Where JavaScript runs
- script/module
- console
- strict mode idea

Practice checkpoint

Explain where javascript runs in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

2. Variables and Types

What to learn

- let/const
- primitive types
- typeof
- null vs undefined
- type conversion

Practice checkpoint

Explain let/const in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

3. Operators and Conditions

What to learn

- comparison
- truthy/falsy
- if/switch
- ternary

Practice checkpoint

Explain comparison in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

4. Loops

What to learn

- for/while
- for...of
- break/continue

Practice checkpoint

Explain for/while in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

5. Functions

What to learn

- declaration/expression
- arrow functions
- parameters/defaults
- return
- scope and closure

Practice checkpoint

Explain declaration/expression in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

6. Arrays

What to learn

- push/pop/map/filter/reduce/find
- spread/rest
- destructuring

Practice checkpoint

Explain push/pop/map/filter/reduce/find in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

7. Objects

What to learn

- properties/methods
- optional chaining
- JSON
- object destructuring

Practice checkpoint

Explain properties/methods in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

8. DOM

What to learn

- querySelector
- textContent
- classList

- createElement
- event listeners

Practice checkpoint

Explain queryselector in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

9. Forms and Validation

What to learn

- submit event
- FormData
- client validation
- error messages

Practice checkpoint

Explain submit event in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

10. Async JavaScript

What to learn

- Promise
- async/await
- fetch
- try/catch
- AbortController

Practice checkpoint

Explain promise in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

11. Modules

What to learn

- export/import
- module scope

Practice checkpoint

Explain export/import in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

12. Storage

What to learn

- localStorage

- sessionStorage
- JSON serialization

Practice checkpoint

Explain localstorage in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

13. Debugging

What to learn

- Console
- breakpoints
- network errors
- common TypeError causes

Practice checkpoint

Explain console in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

14. Project

What to learn

- Build a task/job tracker
- Add filters
- Persist locally
- Fetch data from API

Practice checkpoint

Explain build a task/job tracker in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

15. Interview Preparation

What to learn

- var/let/const
- == vs ===
- closure
- event bubbling
- promise vs async/await
- map vs forEach

Practice checkpoint

Explain var/let/const in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

Worked Examples

Array map

```
const names = users.map(user => user.name);
```

Fetch

```
const response = await fetch(url); if (!response.ok) throw new Error("Request failed");
```

DOM

```
button.addEventListener("click", () => panel.classList.toggle("open"));
```

Interview Answer Pattern

Use this structure for technical questions:

- 1. Definition - say what it is in one or two lines.
- 2. Why - explain why it is useful.
- 3. Example - give a small real or code example.
- 4. Difference/common mistake - compare with a related concept when relevant.
- 5. Be truthful - do not claim tools or experience you have not used.

Final Course Checklist

- I can explain the foundation without notes.
- I completed examples by typing/doing them myself.
- I completed topic tests and reviewed wrong answers.
- I completed at least one mini project or practical exercise.
- I can answer common interview questions in simple language.
- I am ready for the MAGIVOX 50-question final exam.